



Research article

Simulation-based driver scoring and profiling system

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ABSTRACT

This paper describes a rule-based Driver Scoring System model, derived from behavioral data collected using a driving simulator. It introduces a novel approach to establish driver profiles through feature engineering of acquired dataset, with features evaluating various aspects of driver behavior. The research aims to provide employers and drivers with profile-specific feedback and recommendations to design training protocols. Principal Component Analysis is applied on preprocessed dataset from 412 drivers for dimensionality reduction and feature selection. The K-means clustering algorithm is used for data analysis, resulting in three distinct clusters. The Kruskal-Wallis test, supplemented by post hoc Dunn testing is employed to determine statistical significance between clusters. Clusters are portrayed using descriptive statistics, specifically the mean scores and overall driver performance averages. Our method delineates three driver profiles, with two driving styles reflecting desirable driving skills and good overall performance, while the third represents unacceptable driving skills and bad overall performance.

1. Introduction

Driving safety is the paramount topic in on-road public transport as traffic crashes, and, consequently, road traffic injuries still represent the leading cause of death, particularly for children and young people aged between 5 and 29 years [1]. Despite the average annual reduction in the number of road deaths of 19.2 % in 2020, and the fact that the number of road deaths worldwide has fallen 5 % since 2010, current death rates (1.19 million for 2021 [1]) are still unacceptable.

After obtaining a driving license, drivers develop their driving style and habits. The latter strongly depends on individual driving skills and behavior, learning experiences, proneness to stress, cultural environment, as well as on potential crash history [2]. According to Sagberg et al. (2015), driving style refers to "those driving behaviors that have developed into habits and occur consistently across driving occasions" [2]. Therefore, driving style can be defined as the way someone drives and often does not change significantly with the improvement of driving skills or driving experience [3]. In contrast, driving behavior involves actions that occur more occasionally, which may not necessarily become habitual. Driving style reflects conscious choices on how to drive, including personal preferences for specific maneuvers [4]. Common driving styles include: patient and cautious, aggressive, reckless or careless, anxious, and sometimes also distracted driving [3]. On the other hand, driving skill portrays an individual driver's performance in a driving task and how well the driver copes with various challenges related to vehicle maneuvering and information processing requirements [5].

Given the critical role of driving behavior in road safety, the emergence of driver performance scoring systems has proven invaluable [6]. These systems go beyond traditional assessments of driving skills by driving instructors and consider multiple

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information sources needed to better understand an individual's driving style and behavior profiles. By analyzing various aspects of driving behavior, such as adherence to traffic rules [7,8], response to challenging road scenarios [9,10], and the degree of cautiousness or aggressiveness [11], driver scoring systems offer a detailed view of a driver's skills and behavior patterns behind the wheel.

Driver scoring systems have found applications in diverse domains, from insurance to fleet management companies and various road safety initiatives. Insurance companies utilize driver scoring systems to tailor insurance premiums to the individual driving behavior, promote safer practices and incentivize responsible driving [12]. Some examples of such systems are State Farm Mutual Automobile Insurance Company [13], DRAJV Slovenian Car Insurance App [12] and Drive Performance Evaluation for UBI developed by Ituran [14]. Fleet managers, on the other hand, rely on these systems to enhance the safety and efficiency of their operations, and ultimately reduce the risk of crashes and the associated costs [15]. Moreover, road safety initiatives benefit from the insights of these systems to mitigate risk by selecting and hiring drivers based on their performance or providing targeted training programs and other types of interventions for their existing employees [16].

Driver scoring systems not only collect and analyze data crucial for understanding driver behavior, but also provide a powerful platform for driver profiling. Driver profiling involves systematic characterization and grouping of individual driving styles, behaviors, and preferences.

In reviewing existing research on driver behavior assessment, we found three primary methodologies: telemetry-based research, traffic surveillance-based research, and driving simulator-based research.

Telemetry-based research focuses on collecting driving data such as speed, acceleration, throttle position, and other vehicle parameters through On-Board Diagnostics (OBD) and smartphone sensors. For instance Ref. [17], proposed a cloud-based platform to collect OBD data and distinguished between eco-friendly and aggressive drivers using clustering algorithms like K-means. In Ref. [18], smartphone sensor data from over 10,000 trips was analyzed to identify driver behavior patterns, finding that profiles could vary significantly between trips, highlighting the temporal variability of driving styles. Similarly [19], utilized SVM classification to categorize sessions as aggressive or non-aggressive based on parameters like lane deviation and time-to-collision, achieving a high accuracy rate of 93.1 %. The key finding across these studies is that telemetry-based data effectively captures driving patterns in real-time but struggles to evaluate behavior under potentially dangerous conditions due to safety constraints. The primary gap lies in the lack of controlled assessments of how drivers behave under extreme stress or risk, which real-world data cannot safely provide.

Studies in traffic surveillance-based research category utilize video data from traffic cameras to analyze vehicle trajectories and interactions. In Ref. [20], vehicle trajectory data was used to assess rear-end collision risks, categorizing driving behavior as aggressive or non-aggressive. Similarly [21], analyzed driver stopping and approaching patterns at intersections, categorizing behavior through clustering into rule-compliant and aggressive approaches. These studies offer valuable insights into driver behavior in specific real-world contexts, such as intersections or congested areas. However, they rely exclusively on observable data, which limits the depth of behavioral analysis. Without contextual or physiological information, it is challenging to capture underlying intentions or emotional states that may influence driving decisions.

Driving Simulator-based research provides a controlled environment where potentially hazardous situations can be safely replicated. In Ref. [22], driver behavior was studied by analyzing steering, yaw rate, and lateral acceleration, ultimately classifying drivers as cautious, normal, or aggressive. In another study [23], the influence of cell phone use on driving performance was explored, revealing that hand-held phone use increased collision risk and braking response time. These studies demonstrate that driving simulators are effective for assessing responses to distractions and risky conditions without compromising safety. The gap addressed by simulator-based studies is the ability to systematically assess driver behavior under standardized stressful conditions, which cannot be achieved through telemetry or surveillance data alone.

While telemetry-based, surveillance-based, and simulator-based studies provide valuable insights into different aspects of driver behavior, they each have their limitations. Telemetry data offers real-time insights but lacks the ability to assess behavior under dangerous conditions. Surveillance data provides an observational perspective but lacks depth without physiological or contextual data. Simulator-based studies allow for detailed examination of driver reactions under controlled conditions but may not always generalize perfectly to real-world behavior.

Our research leverages the significant advantages of driving simulator-based methodologies by integrating diverse evaluation metrics, including biometric data, within a modular, adaptable, and scalable Decision Support System (DSS). The proposed DSS provides a comprehensive, end-to-end solution for assessing driver behavior across multiple dimensions, offering valuable feedback for both training and evaluation, while maintaining flexibility and applicability to a wide range of use cases. The system's outputs accurately reflect real-world characteristics and traits of individual drivers, ensuring high repeatability, robustness, and ecological validity.

1.1. Aim of the paper & contributions

The primary contributions and aims of our research are as follows.

1. Develop an objective and comprehensive method to assess and classify driver behavior for proactive road safety measures, informed driver hiring processes, and tailored driver training programs.
2. Introduce the Driver Scoring System, a modular, scalable, and adaptable rule-based model derived from behavioral data collected in a driving simulator. The DSS has a modular and adaptable architecture, allowing for the flexible addition or removal of evaluation categories and adjustable weights for different metrics. It evaluates various aspects of driver behavior, such as compliance

with road regulations and reactions in critical situations, within a controlled environment, allowing for comprehensive feedback without real-world risks.

3. Incorporate biometric data, such as heart rate, galvanic skin response, and gaze data, to provide a deeper understanding of driver behavior beyond simple metrics. This comprehensive approach closely reflects real-world conditions and facilitates the development of tailored training programs in both academic and industrial domains [24].
4. Enhance transparency and interpretability in driver scoring by using a rule-based approach instead of neural networks, which often lack explainability [24,25]. This enables the distinction between deliberate recklessness and cautious driving under stress or impairment, contributing to safer driving practices.
5. Employ driver clustering to delineate behavioral patterns and correlate these with performance scores. This clustering approach, based on data from over 500 drivers, helps identify problematic behaviors and supports better driver-job matching, offering deeper insights compared to simpler feature-based clustering [26,22,23].
6. End-to-end solution: The DSS is designed as a complete end-to-end solution for driver assessment, incorporating data collection, analysis, driver profiling, and feedback generation. This comprehensive system supports proactive safety measures and contributes valuable advancements in both scientific research and practical driver evaluation applications.

2. Driver scoring system description

This section provides an overview of the approach and procedures used in the development and implementation of the DSS in conjunction with the driving simulator. It begins by offering details on the hardware and software configuration, scenario customization capabilities, and the user interface. Furthermore, it delves into the design and development process [24] of the DSS, explaining the methodology for data collection and analysis.

2.1. Driving simulator and driving scenarios

The driving simulator consist of a vibration-absorbing dome, a patented proprietary 4DOF motion platform, an adjustable car seat, and a set of standardized vehicle controls such as a steering wheel with direct drive servo force feedback, pedals, a gear selection lever, and screens with instrument cluster and infotainment system. The motion system can effectively simulate acceleration, braking and different road surfaces.

Its display system consists of three 50-inch LCD screens with 120° field of vision. Driver engagement is also enhanced through a 5.1 surround sound system, that produces lifelike engine and vehicle sounds. Three powerful computer systems are used to run the simulation software. The simulation software is based on the SCANer™ Studio environment (AVsimulation, Boulogne-Billancourt, France), which provides comprehensive tools and models for building ultra-realistic virtual worlds, including road environments, vehicle dynamics, traffic, sensors, drivers, headlights, weather conditions, and scenario scripting. It uses advanced Artificial Intelligence (AI) algorithms to generate surrounding traffic with different driving styles, ranging from aggressive to unusual and dangerous behavior. Our addition to the SCANer™ software captures and analyzes the driver's reactions and interactions with the simulated environment, with a particular focus on challenging and critical situations that cannot be easily replicated or trained for in real-life.

Our driving scenario consists of a 20-min ride through three distinct environments: a two-lane highway, a two-way country road, and a bustling city center. These scenarios demand high levels of focus, exceptional driving skills, and swift decision-making to avoid collisions and prevent crashes. Some of these critical situations include unexpected appearance of pedestrians (especially in the city center), aggressive maneuvers of other drivers, unpredictable behavior of motorcyclists, sudden weather changes that reduce visibility (like dense fog) and other. The software is further enhanced by proprietary modules that enable seamless integration with external sensors to assess the driver's psychophysiological state, ensuring a comprehensive evaluation of driver performance.

Each developed driving scenario contains Scenario Settings, a file that contains information on which objects must or must not be seen, their ID, type, description and score (e.g. billboards, traffic signs), defined events when the driver's reaction time is measured (e.g. when a vehicle cuts in front of the driver), defined events when the safety distance is measured (e.g. when a slow vehicle is driving in the fast lane on the highway), and defined events when the speed violation is measured (by a speed radar). This file also contains a list of maneuvers the driver should or should not perform, their ID, description and the positive (or negative) score if the maneuver is performed. Each maneuver belongs to one of the four evaluated aspects of maneuvering (attention, area insurance, proactivity, and non-aggressiveness). Since the scoring ranges can be defined differently for each scenario - for example, to penalize a certain driver behavior more severely in a specific population - the Scenario Settings also contain ranges for each of the sub-scores, the high-level scores, and the final Driver Score. The Scenario Settings file and scenario development jointly make the DSS modular in nature since various situations can be added to the scenario and evaluated from multiple aspects in a very efficient way.

2.2. Biometric measurement

Biometric measurements are a great addition to the driving simulator, enabling the collection of physiological and biometric data, and a more in-depth driver performance analysis. In our case, the simulator is equipped with two external medical-grade sensors that are synchronized with the simulation software.

The most important external sensor is the Tobii Pro Glasses 2 eye tracker [27], which captures the driver's ocular activity in real-time. This eye tracker records the driver's head position and gaze position in the visual scene. With signal processing, information about gaze direction, fixations, saccades, blinking, and pupil activity can be extracted to provide objective insights into human

behavior and attention during the simulation.

Another important sensor is the Empatica E4 wristband [28] which enables continuous measurements of galvanic skin response (GSR), heart rate variability (HRV), electrodermal activity (EDA), skin temperature, and hand movements.

The integration of these biometric sensors enables high-frequency data collection on the driver's gaze position, current object of interest, electrodermal activity, skin temperature, and cardiovascular data. For example, the analysis of the driver's gaze position in relation to the objects on the simulation screen provides insights into their visual attention and whether specific objects, such as vehicles, pedestrians, or traffic signs, were observed or overlooked by the driver.

High arousal can significantly impact driving performance, impairing decision-making and increasing crash risk due to reduced attention and slower reaction times [29]. Research on eye movement analysis supports this, showing that drivers with high arousal, such as anger, often scan a narrower area and rely more on heuristic processing, increasing the likelihood of missing peripheral hazards [29]. Thus, our biometric integration not only enriches the driving simulation data but also helps correlate physiological responses with driving behavior, providing a deeper understanding of how arousal and attention affect performance.

2.3. Data acquisition

Data acquisition is performed through a central data collection software unit which integrates and synchronizes the data streams from the Empatica E4, the Tobii eye tracking system, and the simulation software. The latter contains information about objects on the road – position, type of traffic signalization, trajectories and velocities of all vehicles and pedestrians, and potential collisions between different objects. The acquired data set also contains information about the status of the hardware components, such as pedal pressure, gearbox state and steering wheel angle.

One of the challenges of using a multi-device platform is the fact that all devices record data at different frequencies: simulator at 500–1000 Hz, Tobii eye tracker at 50 Hz, Empatica at 4 Hz, 32 Hz and 64 Hz (depending on the signal). The data from different sources must be synchronized and stored in one central dataset with a sampling frequency of 100Hz by upsampling (data is interpolated or the last valid sample is used) or downsampling (data is averaged over several samples). This is done in the pre-processing step, along with data integrity validation. All signals found in this final synchronized dataset are listed in [Table 4 in the Appendix](#).

2.4. Driver score calculation steps

As explained in the previous sections, during the simulated drive, the central data recording unit captures a multitude of data points pertaining to the ego vehicle's status, its surroundings, and the driver's physiological responses. The ego vehicle and physiological responses directly relate to the driver's performance, while the surroundings and traffic signalization are defined by the driving simulation scenario.

To calculate the overall Driver Score, the collected raw data is fused, synchronized and analyzed, resulting in three key outputs, shown in [Fig. 1](#).

The simple feature calculation stage consists of a set of sub-steps. Within this stage, the presence and column type are verified ([Table 1](#)) to ensure the use of valid data. Considering data logging challenges, such as data package latency, mandatory timestamp

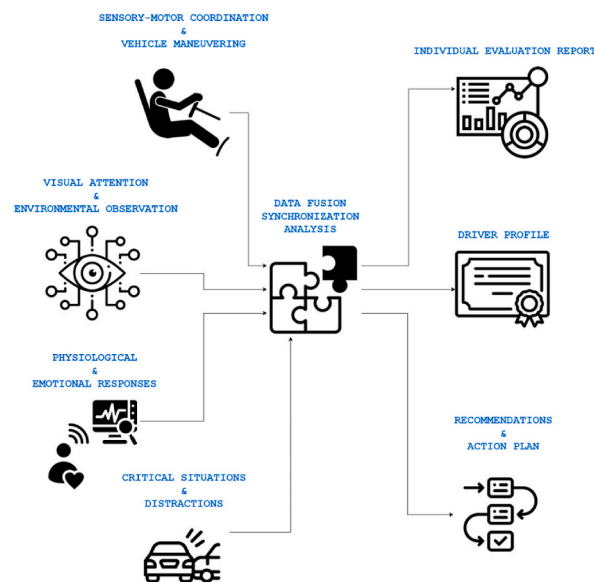


Fig. 1. Driver Score Calculation Workflow - how the system integrates multiple data sources to provide comprehensive and actionable insights for driver evaluation, offering an end-to-end holistic solution.

Table 1
Example weights for different sub-scores and high-level scores (in bold).

SCORE	WEIGHT	VALUE [%]
DEFENSIVENESS SCORE	w1	8.0
SAFETY DISTANCE COMPLIANCE SCORE	w11	33.3
ON ROAD DRIVING SCORE	w12	20.0
SMOOTHNESS SCORE	w13	20.0
SPEED LIMIT COMPLIANCE IN BAD WEATHER	w14	26.7
ECO DRIVING SCORE	w2	3.0
COMPLIANCE SCORE	w3	50.0
CRASH SCORE	w31	33.0
SPEED LIMIT COMPLIANCE SCORE	w32	33.0
TRAFFIC SIGNALIZATION SCORE	w33	33.0
CALMNESS SCORE	w4	3.0
ATTENTION SCORE	w5	8.0
MIRRORS SEEN SCORE	w51	30.0
SCORE OF SEEN OBJECTS	w52	70.0
REACTION SCORE	w6	8.0
MITIGATION SCORE	w7	10.0
MANEUVERING SCORE	w8	10.0
AREA INSURANCE SCORE	w81	25.0
PROACTIVITY SCORE	w82	25.0
ATTENTION SCORE	w83	25.0
NON-AGGRESSIVENESS SCORE	w84	25.0

sorting is performed.

Data cleaning then safeguards data integrity by subjecting the data to a thorough cleaning process. If the drive duration is less than 2.5 min, the analysis is terminated. Missing values are filled in with the most recent known value or interpolated where more appropriate.

The tagging and data segmentation stage prepares and structures the data for further analysis. For example, it calculates and labels instances of speeding (including the percentage of speeding sections), and off-road driving occurrences (0 or 1). The feature calculation stage, on the other hand, provides relevant variables and attributes from raw data by extracting information on lane shifting, time to

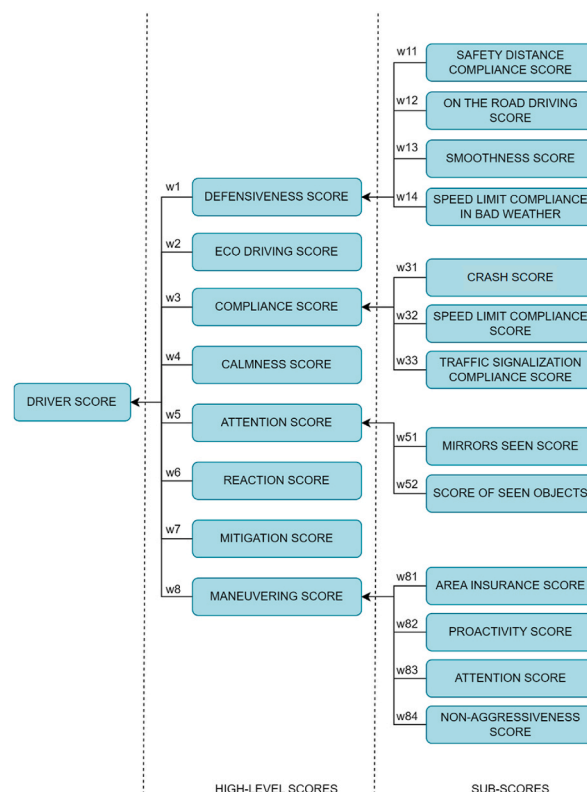


Fig. 2. Graphical representation of the Driver Score, its High-Level Scores and Sub-Scores and corresponding weights (in orange).

collision, frequency of mirror checks and correct intersection detection, off-road gaze situations, and the estimated number of stressful situations.

The analysis stage is crucial as the data is now cleaned and labeled and all features are computed. Sub-scores are calculated to determine the high-level scores that contribute to the overall Driver Score. These sub-scores include aspects such as speed violation in foggy conditions, mirror checking, fines and penalty points, number of collisions, reaction time at critical events, safety distance violation, successful maneuvering, compliant use of vehicle indicators, off-road driving etc., all of which are presented in [Section 2.5](#).

In the post-analysis stage, the sub-scores and then the high-level scores from the analysis stage are weighted and combined linearly. There are eight high-level scores which constitute the final Driver Score: Attention Score, Defensiveness Score, Compliance Score, Calmness Score, Eco-driving Score, Alertness Score, Mitigation Score, and Maneuvering Score.

2.5. Sub-scores, high-level scores, and overall driver score

This section presents each of the high-level scores and the sub-scores used to calculate them, as well as the calculation process of the overall Driver Score. The graphical representation of this hierarchical model can be seen in [Fig. 2](#).

2.5.1. Driver score

The final Driver Score in the DSS is a comprehensive numerical representation (ranging from 0 to 100) of the driver's overall performance. To calculate the Driver Score, a rule-based system is used, providing clear and transparent scoring criteria for each driving aspect.

The final output of this system is a weighted sum of 8 high-level scores, with each score representing a specific aspect of driving performance. This approach ensures that the contribution of each aspect to the final score is understandable and interpretable. The weights assigned to the different scores and categories are determined by optimization methods, such as gradient descent or genetic algorithms, using a pool of drivers with known final scores as a reference, which were obtained with the help of experts and safe driving instructors [25].

2.5.2. Weight determination

The most important goal when determining the final weights is to retain the interpretability of the scoring system. The ability to explain how each category contributes to the final Driver Score remains essential for understanding and validating individual driving performance. To enhance the accuracy and objectivity of rule-based system we initially incorporated neural networks to evaluate and optimize the weights, as reported in our previous work [25]. The study utilized a neural network model with a two-layer architecture: the input layer consisted of 21 neurons, each representing a feature from the rule-based system; the hidden layer had 6 neurons, corresponding to categories in the initial rule-based model; and the output layer had 3 neurons, each representing one of the driver types (safe, average, risky). We compared this neural network model to another model with randomly initialized weights and a Naive Bayes classifier.

The training data included performance data from 37 drivers, collected from a driving simulator that recorded both driving behavior and physiological responses. The ground truth dataset was created by evaluating and scoring the driving performance of each participant through video recordings made with an eye tracker. These recordings captured the participants' point of view and gaze and were assessed by professional driving instructors using a standard form for assessing skills and behavior [30]. The instructors classified the participants into three driver types: risky, average, or safe, based on their evaluations and additional inputs, such as past driving records, crashes, and tickets. The neural network demonstrated improved classification accuracy and precision over the rule-based system, achieving 66 % accuracy compared to 43 % for the rule-based model, and outperformed the Naive Bayes classifier, which achieved 50 % accuracy. While machine learning techniques can enhance accuracy in the initial weight determination process, they often introduce a loss of transparency, limiting their applicability across all potential use cases. For example, in the case of a delivery driver, who must navigate urban environments efficiently and under time pressure, the system can adjust the weights to favor more relevant driving behaviors. This might include increasing the weight for the Maneuvering Proactivity Score which rewards drivers for taking initiative in dynamic traffic conditions, and the Maneuvering Area Insurance Score, which measures risk management in specific areas. At the same time, weights for scores like Smoothness Score and Maneuvering Non-Aggressiveness Score may be reduced, as these drivers might need to prioritize efficiency over comfort.

On the other hand, for a driver of luxury chauffeur service, the focus shifts towards maximizing passenger comfort and safety. In this scenario, the system can increase the weight for the Smoothness Score to reward drivers who ensure a comfortable ride, as well as the Maneuvering Non-Aggressiveness Score, which would penalize any aggressive driving behavior that could disturb the passenger. Additionally, the Attention Score might be emphasized to ensure that the driver is consistently aware of their surroundings and potential hazards, providing a safer and more pleasant experience for the passenger.

By adjusting the weights, the DSS becomes adaptable to various use cases, rewarding or penalizing behaviors based on the specific requirements of different driving roles. This adaptability not only enhances the system's versatility but also ensures that the evaluation remains relevant and aligned with the needs of different stakeholders, such as fleet managers, insurance companies, or individual drivers. This approach allows the DSS to be fine-tuned to match different driving contexts, making it a robust, adaptable, and scalable solution for driver assessment.

The following table provides an example of the weights used in the scoring system. It is important to note that these weights are illustrative since they represent the basic example and can be adjusted according to specific use cases to ensure that the DSS is tailored to the requirements of different driving roles.

2.5.3. Defensiveness Score

The Defensiveness Score is calculated using a formula that takes into account various driving behaviors associated with aggressiveness. Each behavior is assigned a weight (w) that reflects its significance in determining the final grade of defensiveness. The formula is as follows:

$$\text{defensiveness} = w_1 \times \text{safety distance compliance score} + w_2 \times \text{on the road driving} + w_3 \times \text{smoothness score} + w_4 \times \text{speed limit compliance in bad weather}$$

Each weight (w) is determined empirically with the help of experts, such as safe driving instructors, and insights from professional and research articles on road and traffic regulations. The higher the Defensiveness Score, the more cautious and non-aggressive the driving behavior is considered to be. Conversely, a lower score indicates a more aggressive driving style.

The Safety Distance Compliance Score provides valuable insights into a driver's ability to maintain a safe following distance. It considers both specific sections of a route where safety distance is particularly important and an average safety distance in the entire driving route. Only sections where the speed is below a certain threshold are excluded (e.g., in traffic jams). The calculation involves determining the optimal safety distance based on the vehicle speed, taking into account the reaction time of 2 s [31]. The optimal safety distance is calculated as the defined reaction time multiplied by the speed in m/s. Finally, the processing unit assesses the percentage of time in which the safety distance was violated. A higher percentage indicates better compliance with the safety distance, while a lower percentage suggests a greater violation of the safety distance.

The On the Road Driving Score provides an assessment of the driver's ability to stay within the designated driving areas expressed in percentage of time. A higher percentage indicates better adherence to staying on the road, while a lower percentage suggests a greater tendency to deviate from the designated path. Off-road driving is penalized. This calculation takes into account the information about the type of road, such as paved road, sidewalk, freeway lane, or bank, and the time the driver spent deviating from the designated path in relation to the duration of the whole drive.

The Smoothness Score provides valuable insights into the driver's ability to maintain a smooth and comfortable acceleration and deceleration profile. This score is determined based on the acceleration and deceleration rates of the driver and analysis of the acceleration derivative, also known as jerk. Jerk serves as an indicator of driving comfort, with lower values indicating a smoother driving experience.

If the jerk remains consistently below a certain threshold, the ride is considered acceptable and comfortable. At speeds between 100 and 130 km/h (standard limits on highways), driving is considered uncomfortable if the absolute jerk is $0.3 - 0.9 \frac{m}{s^3}$ [32]. The longitudinal jerk above $1.07 \frac{m}{s^3}$ is typically associated with aggressive acceleration, while values below $-1.43 \frac{m}{s^3}$, are associated with aggressive deceleration [33].

The Speed Compliance in Bad Weather Score is calculated in areas where visibility is less than 400 m, and it is based on [34]. It emphasizes the importance of adhering to appropriate speed limits to increase safety and avoid crashes in challenging weather situations. The key concept is to ensure that the driver's stopping distance in fog is less than the current visibility. The stopping distance is calculated as the sum of the reaction time distance, which is the distance the driver would have covered during the reaction time (set at 2 s), and the braking distance.

The braking distance is determined using the formula $d_{\text{braking}} = \frac{\text{speed}^2}{2 \cdot g \cdot k_{\text{friction}}}$ where speed is the driver's current speed, g is the gravitational acceleration, and k_{friction} is the friction coefficient, with values typically ranging from 0.2 to 0.7 (depending on the brake manufacturer and the type of pavement surface defined in the scenario). By comparing the total stopping distance with the current visibility, the speed limit violation in bad weather is assessed. If the driver's speed exceeds the calculated stopping distance, it indicates a violation of the speed limit, posing a safety risk in foggy/bad weather conditions with reduced visibility. This assessment intended to encourage drivers to adjust their speed according to the prevailing weather conditions, ensuring they can safely come to a stop within the available visibility range.

2.5.4. Eco-driving score

The Eco-driving Score encourages drivers to adopt fuel-efficient and environmentally friendly driving behavior including smooth acceleration, maintaining moderate speed, and efficient braking. By receiving feedback on their eco-driving performance, drivers can strive to reduce fuel consumption and minimize their environmental impact during the driving simulation.

The Eco-driving Score is determined based on fuel consumption data obtained from the simulator throughout the entire ride. The calculation involves analyzing the data on the current fuel consumption and the distance traveled, allowing the system to calculate the average consumption for the entire journey. Once the average consumption is determined, it is categorized in consumption ranges.

The thresholds for categorizing fuel consumption into consumption ranges are determined based on the specific characteristics of the car used as the ego vehicle in the simulation. To establish these thresholds, a series of test drives are conducted using the same vehicle. During these test drives, detailed fuel consumption data is collected under various driving conditions.

By performing a distribution analysis, identify natural breakpoints using specific percentiles, that represent the points in the distribution where fuel consumption significantly changes. These thresholds are therefore tailored to the specific vehicle used in the simulation, ensuring that the Eco-driving Score accurately reflects the vehicle's typical fuel consumption patterns. As a result, drivers are evaluated against realistic and context-appropriate benchmarks, making the Eco-driving Score both fair and relevant to the specific driving scenario.

Based on the categorized fuel consumption, the system calculates the driver's eco-driving rating. A driver with a lower average fuel

consumption, falling within the "low consumption" range, receives a higher Eco-driving Score, indicating an eco-friendlier driving style. Conversely, a driver with a higher fuel consumption that falls into the "high consumption" range, receives a lower Eco-driving Score, suggesting a less eco-conscious driving style.

2.5.5. Compliance Score

The Compliance Score is calculated based on the evaluation of driving errors and violations of road and traffic regulations. The score is obtained using the following formula:

$$\text{compliance} = w_1 \times \text{speed limit compliance} + w_2 \times \text{traffic sign compliance} + w_3 \times \text{accidents score}$$

In this formula, each w represents a weight that determines the importance of a specific category for the final Compliance Score. These weights are determined empirically with the assistance of experts, such as safe driving instructors, and are based on findings from professional research articles and well-known road and traffic regulations.

By combining these individual sub-scores with their respective weights, the Compliance Score reflects the driver's overall adherence to road and traffic regulations during the simulated drive. A higher Compliance Score indicates better compliance, while a lower score suggests a need for improvement in compliance with road and traffic regulations.

The Speed Limit Compliance Score is determined by assessing the driver's compliance with the speed limit regulations during the simulated drive, measured as a percentage of the time the driver adhered to the current speed limit. The severity of the violation corresponds to the number of penalty points the driver receives. Additionally, fines are also calculated based on the speed limit and the speed of the vehicle, taking into account the radar events specified in the Scenario Settings. The rules for determining penalty points and fines are country-specific, and are also defined in the Scenario Settings.

The Traffic Signalization Compliance Score is determined by evaluating the driver's compliance with traffic signalization expressed as a percentage of violated traffic signs, including stop signs, yield signs, and red lights, individually.

Finally, the Collision Score is calculated based on the driver's involvement in collisions during the simulated drive, expressed as the number of collisions caused by the driver, excluding collisions for which the driver was not responsible.

2.5.6. Calmness Score

The Calmness Score estimates the severity and frequency of stressful situations experienced during the simulated drive by processing the driver's physiological signals. A higher Calmness Score indicates that the driver remained relatively calm and composed throughout the drive, while a lower score indicates that the driver experienced higher arousal levels. The goal is to assess the driver's ability to remain composed and manage arousal in different driving scenarios, providing valuable insights into their overall driving performance and adaptability.

The arousal levels of the driver are calculated from skin conductivity and heart rate. The segment-average heart rate is compared to the driver's median heart rate multiplied by a coefficient (defined using multiple sources [35,36]) to determine whether arousal is detected heart rate-wise. The number of sudden, steep changes in skin conductance are detected using a fusion of different approaches [37–39]. For each segment, the presence of arousal is detected using heart rate and skin conductance. If both modalities indicate arousal, the driver is considered to be under high arousal, otherwise, under mild arousal if either one or the other modality has detected arousal. The detected arousal states are compared with Scenario Settings-defined driving situations, and if the arousal coincided with critical situations defined in the Scenario Settings, the driving segment is displayed to the driver.

2.5.7. Attention Score

The Attention Score for the driver is evaluated based on various aspects of their attentiveness during the simulation. The first aspect is the frequency of checking the rear-view mirrors using the eye-tracking data. The system compares the direction of the driver's gaze with the positions of the mirrors and determines whether the driver has looked at the respective mirror. The percentage of successful mirror checks is then calculated based on the expected interval of mirror checking which is 8 s [40].

The second aspect of the driver's attention is the observation of important objects in the environment and their ability to ignore distractions while driving. Throughout the entire drive, the positions of objects that are relevant for driving and those that could distract the driver are defined in the Scenario Settings. The algorithm compares this data with the driver's gaze directions recorded during the simulation and determines whether they noticed all important objects and avoided the distractions. Paying attention to important objects increases this score while paying attention to different distracting elements decreases it. All positive and negative contributions are finally aggregated using the following formula:

$$\text{attention} = w_1 \times \text{mirrors seen} + w_2 \times \text{score of seen objects}$$

This score provides a comprehensive evaluation of the driver's attentiveness and ability to focus on critical aspects of driving.

2.5.8. Reaction Score

The Reaction Score reflects the driver's ability to respond promptly to critical situations, with a lower average reaction time indicating faster and more efficient reactions. During the simulation, predetermined (Scenario Settings-defined) events are designed to trigger the driver's reaction, and the reaction time is measured.

For instance, one such event is when a football suddenly appears on the road and the driver is required to brake abruptly. The reaction time is measured as: $t_d = t_2 - t_1$, where t_1 is the moment of event occurrence, and t_2 is the moment of driver's reaction, in this

example the moment when the driver pressed the brake. Finally, our algorithm calculates the average reaction time for all such events. A driver who exhibits shorter reaction times receives a better score in the Reaction Score evaluation.

2.5.9. Mitigation Score

The Mitigation Score is a measure of a driver's performance in avoiding risky or dangerous driving behavior and responding appropriately to critical situations. The assessment process begins by identifying Scenario Settings-specified segments of the drive where the driver encounters vehicles ahead or certain events that require a reaction time measurement. These events are used to assess the driver's ability to mitigate potential hazards. The segments are further analyzed for longitudinal and lateral violations, as well as for distance behavior. Some of these events include sudden fog, vehicle cutting in front of the driver's car, approaching an intersection, etc.

The Mitigation Score is divided into two levels of violation, each representing different degrees of severity. The first level includes hard braking, slight braking, or intention to change lanes. The second level includes actions like tailgating and increasing the distance between vehicles.

The final Mitigation Score is calculated as the percentage of successfully mitigated risky behaviors and critical situations. A higher Mitigation Score indicates a more skilled and cautious driver who is adept at responding appropriately to various driving scenarios, thereby minimizing the risk of crashes or unsafe driving actions.

2.5.10. Maneuvering Score

The Maneuvering Score is a comprehensive assessment of individual driving maneuvers in road traffic. A higher score indicates a skillful and responsible driver who performs maneuvers with caution and consideration for others and pays attention to the changing road conditions. Conversely, a lower score may indicate a need for improvement in maneuvering abilities and road awareness. The Maneuvering Score plays a vital role in providing drivers with feedback on their maneuvering skills and promoting safe driving practices on the road.

For each maneuver, specific points of focus and required reactions are defined in advance in the Scenario Settings. The driver's ability to adhere to these predefined situations is evaluated through four key sub-scores: Area Insurance, Non-aggressiveness, Proactivity, and Attention. Each score evaluates a specific aspect of the driver's maneuvering skills and behavior.

Area Insurance considers whether the driver takes the necessary precautions and exercises caution when making lane changes or turns, to ensure the safety of other drivers. Non-aggressiveness assesses how calmly and composedly the driver interacts with other road users. For example, a driver is deemed more aggressive in maneuvering when they honk or flash the headlights immediately after the green light comes on. Drivers who display a patient and non-confrontational attitude receive higher scores in this category. The proactivity aspect evaluates how the driver deals with unexpected situations on the road (e.g., signaling to let the child know they can cross the street, honking/flashing lights if the vehicle in front is not moving for a few seconds, etc.) Drivers who demonstrate foresight and readiness for potential challenges receive higher scores in this category. The maneuvering attention component rewards the driver for paying attention to other traffic participants while maneuvering (e.g., noticing a child between the parked cars, a woman running towards the bus, a truck in the slow lane, etc.) and when they effectively communicate with other road users.

By combining these four components, the overall maneuvers score is evaluated as follows:

$$\text{maneuvering score} = w_1 \times \text{area insurance} + w_2 \times \text{non-aggressiveness} + w_3 \times \text{proactivity} + w_4 \times \text{maneuvering attention}.$$

3. Cluster analysis

This section describes a user study that was conducted to present the DSS in practice and analyze its results. It discusses the methods and steps taken to carefully evaluate how well the system measures driver performance in a simulated scenario. This includes gathering data, clustering drivers based on their high-level scores and sub-scores that make up the Driver Score, analyzing the Driver Score distribution for each cluster, and exploring the relationship between each cluster and driver behavior.

Most importantly, we want to demonstrate that diverse performance scores mirror corresponding driving behaviors and to showcase the system's capability to delineate distinct profiles of professional drivers based on these scores, as a part of the system's



Fig. 3. Setup of the experiment with equipment used for data acquisition in a driving simulator.

final processing step.

3.1. Data collection

The dataset used for the evaluation of the proposed scoring system was collected through the AIDMG [41] in the United Arab Emirates.

The experimental setup included the proprietary compact driving simulator (Section 2.1) which offers a highly immersive and realistic driving experience (Fig. 3). The participants were professional drivers from various transport and delivery companies recruited as a part of the project. The user ID (user number, driving scenario, take number) and demographic data - age (39.5 ± 9.28 years), driver's license, company, nationality, and gender - were also part of the dataset (possible values are listed in Table 2). For this study, we collected 538 entries of individual driving sessions. The study was conducted in line with the general principles outlined in the Declaration of Helsinki to ensure that ethical standards and confidentiality of the data were maintained. Since the trials were carried out as part of routine job training rather than explicitly for experimental purposes, a formal ethics committee review was not required.

Participants were required to engage in an approximately 20-min driving session encompassing three distinct driving environments: a two-lane highway, a two-way rural road, and a bustling city center. This is a standardized driving scenario required for the scoring system with some modifications tailored to the Middle East driving environment.

Additionally, the compilation of the final list of critical situations in the scenario was guided also by their significance in achieving the objectives of the driver profiling, provided by professional driving instructors. It is important to add that the modular nature of the Driver Score presented in this paper allows for customization of the driving Scenario Settings while keeping the data collection process consistent. For each of the 538 driver entries, a vector comprising the overall Driver Score and 19 high-level scores and sub-scores described in Section 2.5 was generated, resulting in a total number of $538 \times 20 = 10760$ features for the entire data set.

3.2. Driver clusterization

The second goal of this research was to present the different driver profiles identified in the final stage of the DSS - clustering on the calculated individual scores and the process used to obtain them. Due to the multicollinearity of the features of DSS architecture, we decided to first use the Principal Component Analysis (PCA) as a feature reduction technique. To prepare the data for PCA, four categorical features out of the total 19 were excluded: Speed Limit Compliance Score, Traffic Signs Compliance Score, Crash Score, and Maneuvering Non-aggressiveness Score. We also had to exclude the instances with missing values, mainly due to incomplete DS vectors. The most problematic categories were Attention and Reaction scores, which accounted for 6 % and 11 % of instances, respectively. These two preprocessing steps resulted in 15 standardized features from 412 observations, yielding a total of 6180 features (15×412). These 15 standardized features were then used as input for PCA.

The optimal number of principal components was determined using a scree plot which arranges the eigenvalues in descending order. The threshold for the cumulative variance was determined by compromising, in order to achieve balance between reducing the dimensionality of the data and preserving the most essential information.

To ascertain classification criteria in the absence of labeled drivers, an unsupervised learning method - K-means was deployed. Unlike supervised learning methods that require labeled data for training, K-means is an unsupervised learning algorithm that can group similar data points based solely on their features. This makes it particularly suitable for our analysis, where we want to identify distinct driver profiles without the need for labeled driver data. Additionally, K-means is computationally efficient and scalable, so it can be applied to large datasets such as ours [42].

At its core, K-means begins by selecting an a priori number of clusters, denoted K . These clusters are represented by centroids, initially chosen either randomly or by specific strategies. The algorithm then proceeds through an iterative process in which the data points are assigned to the nearest cluster centroid based on a chosen distance metric, usually Euclidean distance [43].

The result of the K-means clustering process is a partitioning of the dataset into K distinct clusters, each characterized by a centroid that represents the "center" of that cluster. This enables the exploration of inherent structure of the data, identifying groupings of similar data points, and gaining insights into the underlying patterns or classes within the dataset. Despite being computationally demanding for large-scale datasets, when used on low-dimensional data K-means is an efficient algorithm, so the use of principal components as a preprocessing method is recommended [44].

The optimal number of clusters in K-means clustering is typically determined by using the Elbow method [43]. This involves evaluating the percentage change in the number of clusters and their associated Sum of Squared Errors (SSE). Essentially, the Elbow

Table 2
The list of demographic data attributes and their possible values.

Property	Possible values
Age	25–45 years
Driver's License	unique ID
Company Type	security, public transport, delivery, logistics
Nationality	Pakistan, India, Lebanon, Nepal, Syria, Yemen, Egypt, Bangladesh, Philippines, Nigeria, Palestine, Ethiopia, Uganda
Gender	male or female

Method helps to determine the cluster number at which the data starts to cluster effectively, and further division does not significantly improve the clustering quality.

In the final step we also performed a statistical analysis to determine whether there were significant differences between the high-level scores of the different clusters, with the objective of determining whether one cluster statistically dominated the others in terms of these scores. We used the Kruskal-Wallis test [45] to identify the presence of statistical differences between the clusters and the Dunn test [46] for non-parametric pairwise multiple comparisons in the second step. This combined approach of Kruskal-Wallis and Dunn tests allowed not only the identification of significant differences, but also the determination of which cluster(s) showed dominance in the context of high-level scores.

For all analysis steps we used Python version 3.9.7 (Python Software Foundation, Delaware, USA).

4. Results

The final set of 412 driver scores represents a statistically relevant sample size that allows us to observe population-specific performance in each driving aspect and examine its anomalies for future redesign and improvements.

The objective of the clustering analysis was to examine the scores generated by the DSS and to identify distinct driver profiles based on these scores. Due to the multicollinearity of the features, we performed dimensionality reduction using PCA before applying the K-means clustering algorithm to the data.

4.1. Dimensionality reduction results

The optimal number of principal components used as input for the clustering algorithm is determined by the proportion of variation explained by each principal component and the eigenvalues (ordered from largest to smallest).

The Scree Plot shown in Fig. 4 shows the amount of variance explained by the individual components. The orange line in the graph represents the cumulative contribution to the variance. In our case we aimed for an optimal balance between dimensionality reduction and preservation of essential information in the data, we decided to keep six PCs, which together contribute to a satisfactory 70 % [47] of the cumulative variance.

In Fig. 5. The spider plot visually summarizes the connections between PCs and high scores. The plot highlights the relative impact of each high score on the PCs and helps to interpret the importance of the PCs in this analysis. For example, the Defensiveness Score has the greatest influence on PC1, the Maneuvering Score on PC2, the Attention Score on PC3 and PC4, the Mitigation Score on PC5, and the Reaction Score on PC6.

4.2. K-means results

Fig. 6 shows the Within-Cluster Sum of Squares (WCSS) for different numbers of clusters. The 'elbow' point is not marked in the graph as it lies somewhere between three and four. Since this point should represent the balance between maximizing clustering effectiveness and avoiding over-segmentation, and there is no significant difference between the WCSS for three and four clusters, we decided to set the cluster number to three.

After performing the clustering process using 6 chosen PCs as K-means input, the results are first visualized in 2D and 3D spaces, using PC1, PC2, and PC3, since the first two contain 37 % of the retained information and the first three contain 47 % of retained information. Cluster separability is also visible when only PC1 and PC2 are used, which contain 37 % of the retained information, proving that in this case dimensionality reduction not only removes redundant features (scores) but also improves cluster separability. Fig. 7 shows the 2D scatter plot for PC1 and PC2 (shown on the left), while a 3D scatter plot for PCs 1–3 with cluster separations is shown on the right.

To describe the three resulting driver profiles (assuming that each cluster represents a driver profile), the feature mean values for each cluster are displayed in Fig. 8. The expectation that a driver profile corresponds to a cluster arises from the assumption that

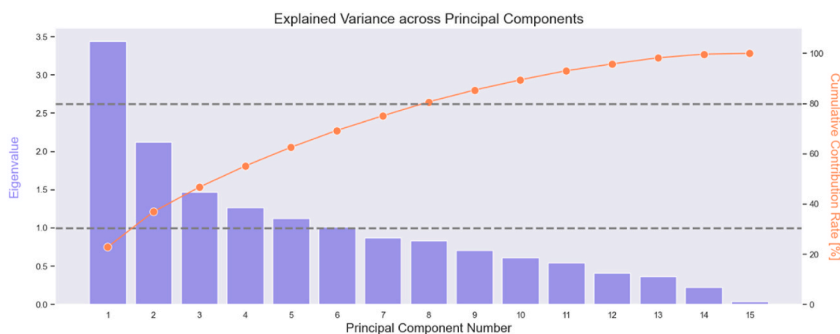


Fig. 4. Scree Plot showing Eigenvalues and Cumulative Variance explained by Principal Components. The figure displays the eigenvalues (ordered from largest to smallest, purple) and the cumulative variability explained by each principal component (orange).



Fig. 5. The spider plot illustrates the relationships between principal components (PCs) and high-level scores. Each PC is represented by a different colored line, and its correlation with specific scores is highlighted. This visualization provides insights into which scores have the most considerable influence on each PC.

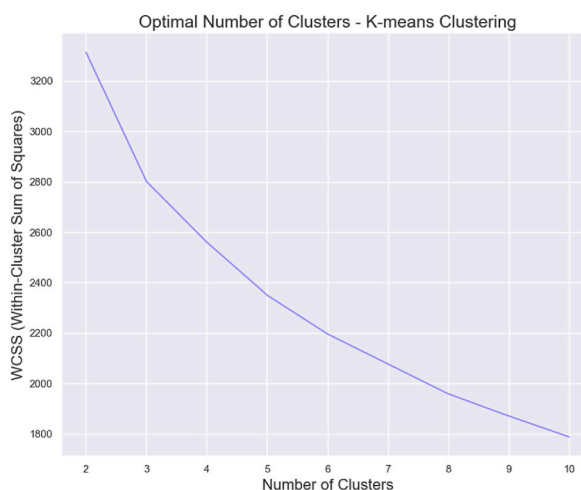


Fig. 6. Elbow Method: Determining optimal number of clusters for K-means clustering algorithm.

different driving styles or behaviors can be characterized by distinct patterns within the scoring system. Clustering algorithms aim to group similar data points based on certain features, in this case a set of scores reflecting various driving aspects.

If the clustering algorithm works effectively, it should identify groups of drivers who have similar patterns in their scores. The rationale is that drivers within a cluster share commonality in their driving behavior, that distinguishes them from drivers in other clusters.

Fig. 8 also shows that there is apparent inter-cluster variability of high-level score means. Cluster 1 has the highest mean for Attention Score, Eco Driving Score, Compliance Score, and Maneuvering Score. Cluster 2 has the highest mean for the Calmness Score and the lowest mean for the Compliance Score, Eco driving Score, and Maneuvering Score. Cluster 3 has the highest Defensiveness and Mitigation Score and the lowest Calmness and Alertness Score.

The Kruskal-Wallis test showed that for each high score, there are at least two statistically significant clusters. The post-hoc Dunn test identified the exact pairs of clusters for which there is a statistically significant difference between the high score means. **Table 3** shows the p-values with a 95 % confidence interval for all comparisons. The bolded p-values indicate a statistically significant difference between the high scores of the clusters.

5. Discussion

By interpreting all the related clusterization results, it can be deduced that Cluster 1 contains the drivers who pay the most attention to the road and signalization and are fairly attentive and good at maneuvering but express aggressive traits and insufficient mitigation. Even though these drivers have a relatively high final DS mean, they have a non-defensive way of driving which includes violating safety distance and jerky driving – probably why they have a tough time mitigating risky situations which lead to hard braking,

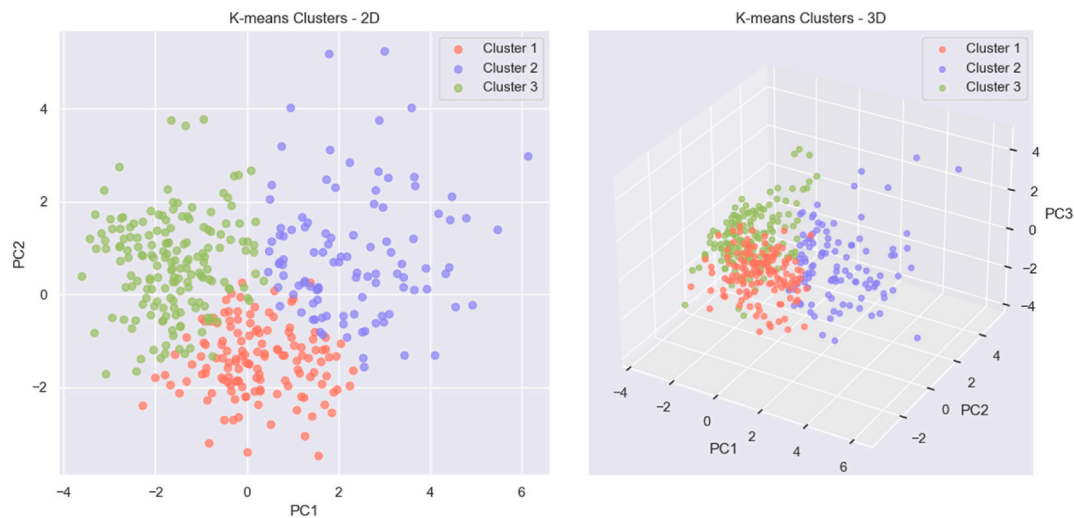


Fig. 7. Two-dimensional and three-dimensional K-means cluster visualization. 2D scatter plot of PC1 and PC2 for (left) and 3D scatter plot of PC1, PC2 and PC3 (right). The sample color is determined by its cluster. PC stands for Principal Component.



Fig. 8. Grouped bar plot of high-level score mean values for K-means clustering algorithm. Bar color is determined by its cluster.

Table 3

Pairwise results of the Kruskal-Wallis and Dunn test displayed using the p-values. A 95 % confidence interval is used, and the bolded p-values indicate that there is a statistically significant difference between the high scores of clusters C_i and C_j : $(i, j) = \{(1, 2), (1, 3), (2, 3)\}$.

Main score	C1-C2	C1-C3	C2-C3
Driver Score	<.001	0.669	<.001
Attention Score	<.001	<.001	0.621
Defensiveness Score	0.646	<.001	<.001
Compliance Score	<.001	0.312	<.001
Eco-driving Score	<.001	0.178	<.001
Calmness Score	.003	0.103	<.001
Alertness Score	0.081	<.001	.002
Mitigation Score	0.006	.002	0.987
Maneuvering Score	<.001	<.001	<.001

tailgating, lane switching, etc. They do not have a low Compliance Score because they are attentive and alert while driving, allowing them to react appropriately.

Cluster 3 has a similar DS mean to Cluster 1 (not entirely significantly different, see Table 3), but a very defensive and compliant way of driving, with good mitigation and maneuvering skills. The downside these drivers express is their low alertness, i.e., long reaction time, which is probably the reason why they also have a high stress level (low Calmness Score), and that is probably a price to

pay for being fully compliant and defensive. Trying to think of the best way to react, mitigate risky situations and how to perform a certain maneuver certainly prolongs reaction time, especially in inexperienced drivers and raises their stress level. Fig. 9 clearly shows that Cluster 2 has the lowest DS mean, while Cluster 1 and Cluster 3 have comparable DS means but some of their other scores are different with a statistical significance (Table 3) ultimately making them separable.

The last and the worst cluster (in terms of driver performance) is Cluster 2, with the lowest DS mean. It contains the most aggressive and non-compliant drivers; their fuel consumption is usually very high due to the heavy braking, jerky drive, and safety distance violation. In addition, they often violate the speed limit in bad weather and spend too much time driving off the road. Their Maneuvering Score is the lowest among clusters, but, not surprisingly, their Calmness Score is the best among clusters, probably because this violent way of driving does not seem problematic to them, hence their style of driving.

These differences suggest that there are distinct groups of drivers exhibiting variations in driving behavior as captured by high scores. From a practical standpoint, this could enable.

1. risk profiling - if one cluster consistently exhibits lower Defensiveness and Compliance Score it could prompt targeted safety measures for that cluster; drivers belonging to that cluster could also be flagged as liable during the driver recruitment process.
2. personalized feedback – drivers within each cluster may benefit from personalized feedback based on the specific aspects where their driving behavior differs significantly from other clusters, etc.
3. foundation for continuous improvement – as more data is collected over time, clustering analyses can be re-run to adapt to the changes in driver behavior patterns and provide scoring system enhancement.

The three resulting driver profiles or classes presented in this paper could be compared to those described in Ref. [48] but only to some extent. They described their classes as “conservative,” “calm,” and “radical.” It can be argued that their “radical” class corresponds to the “Bad drivers” class, while the “Non-defensive” and “Compliant” drivers correspond to a combination of “conservative” and “calm” drivers. The drivers analyzed in this paper generally exhibited a higher level of aggressiveness and less calmness (but still compliant with traffic rules) as they were all professional drivers typically required to be efficient and flexible in dense traffic situations.

Comparing these resulting three driver profiles to six clusters obtained in Ref. [18], it can be observed that similar driving aspects indicate if a driver is less or more aggressive and if their driving is considered safe. In Ref. [18], acceleration and deceleration profiles with abrupt changes, speeding and phone usage indicate unsafe drivers, similarly to a low Defensiveness, Compliance and Attention Score in case of the three clusters obtained in our study.

The low (exhibited by “Bad drivers” class) and high (exhibited by other two classes) Eco-driving Score coincide with the results from Ref. [49] where drivers were separated into eco-friendly and pollutive (aggressive) clusters based on their speed and RPM, both of which largely influence the fuel consumption, based on which the Eco-driving Score is defined.

5.1. Study limitations

The following potential improvements and limitations were recognized in the methodology and results.

1. Future work could involve obtaining a more diverse and larger sample to explore the possibility of forming more granular clusters with specific attributes. It could also investigate whether the same data model is applicable in different driving simulation scenarios.
2. Missing features led to the removal of data entries from the analysis because even though their DS can be calculated, there is no straightforward way of imputing a feature that is typically calculated by using the data from the whole drive. In cases where certain features are missing, their absence hinders the clear determination of a driver profile.
3. Driver fatigue is a crucial factor, particularly in professional drivers who experience daily fatigue due to their work nature. To comprehensively analyze driver profiles, future work should consider testing drivers at specific times [50] in their shifts or conducting longer drives to induce and monitor the influence of fatigue [51] on driver profiles.

6. Conclusion

In this study, a DSS is presented, and a method is proposed to classify drivers into specific clusters, i.e., assigning them a driver profile after they have been tested in a driving simulation test, with the goal of providing relevant feedback within their training program and outlining ways to improve their performance. Clustering results and cluster-specific high score means suggest that the DSS put in place has great potential and could help determine the driver's profile. Furthermore, even if there is no clear border between clusters (driver profiles) – which is somewhat expected since some drivers can have characteristics of two or more profiles - there is a statistically significant difference between certain cluster-specific high score means which makes it possible to define three different clusters with the following names: Non-defensive drivers (Cluster 1), Compliant drivers (Cluster 3) and Bad drivers (Cluster 2).

In summary, clustering results of our study align with findings from various studies, offering a multi-faceted understanding of driver performance. The distinct driver profiles identified in this research, characterized by variations in defensiveness, compliance, attention, maneuvering etc., align with and extend existing research by correlating certain driving aspects with aggressiveness, safety, and eco-friendliness – not separately, but all at the same time.

The ability to assign a specific driving profile to an individual driver could provide an opportunity to define concrete feedback and

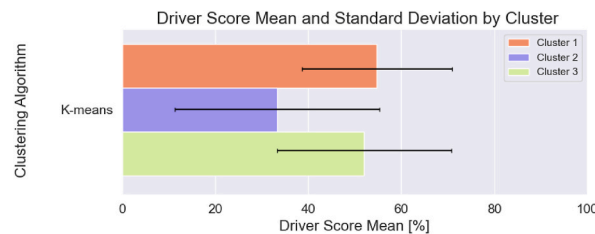


Fig. 9. Driver Score comparison for both clustering algorithms (K-means and Agglomerative Clustering) and for all clusters.

recommendations for improving their driving skills. It could also improve the screening process for hiring new drivers in companies where the worst drivers would be rejected while drivers of the other two clusters would be assigned to different driving jobs (e.g., driving people, goods, fast mail, etc.)

Furthermore, this data-driven decision-making process opens avenues for the design of tailored training scenarios for different driver profiles. These scenarios can focus on improving the performance of drivers with subpar scores, providing them with opportunities to enhance their skills and attain a better driver profile. These are just a few potential implications of this approach.

Beyond the theoretical concepts, our system has demonstrated its practical utility in real-world applications. It is currently being used in Zavarovalnica Triglav's insurance premium definer for young drivers, where those who perform well on the test can receive a discount on their premiums, therefore directly linking their performance to tangible financial benefits. Additionally, the system is employed in the evaluation of Dubai Police officers in a specialized driving hub, ensuring high standards in their driving skills. Lastly, it plays a crucial role in driver recruitment [41] by acting as an initial skill assessment tool, helping employers by filtering out candidates who do not perform well, thereby saving time and resources on insurance and employment processes for candidates who may not meet the necessary driving performance standards, set by each employer to its own preferences. These applications highlight the system's versatility, effectiveness and practicality.

CRediT authorship contribution statement

Jelena Medarević: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Sašo Tomažič:** Writing – review & editing, Funding acquisition. **Jaka Sodnik:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data and code Availability

The data and code used in this study are not publicly available due to proprietary restrictions. However, the data can be made available from the corresponding author upon reasonable request for research purposes. Interested researchers are encouraged to reach out for access.

Ethics statement

This study was conducted in accordance with the principles outlined in the Declaration of Helsinki.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to assist with revising and improving the grammar and clarity of the text. After using this tool, the authors reviewed and edited the content as necessary and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table 4

The list of main raw signals collected during the driving session, along with simple features calculated during the first step of the analysis.

Parameter	Description
USER	
USER_ID	Number representing user identifier.
TASK	
SUBTASK_NUMBER	Number that indicates which subtask is currently active.
SUBTASK_ACTION	Number that indicates what activities are performed during the subtask.
PHONE_CALL	Number that indicates if user is using a mobile phone while driving (0 – no phone usage, 1 – active phone usage)
TIME	
DATE	Date of the start of the simulation in YYYY-MM-DD-H:M:S:MS format.
TIMESTAMP	Current UNIX timestamp.
TIME_OF_DRIVING	Time elapsed from the start of scenario, in seconds.
INTERACTIVE VEHICLE PARAMETERS	
POSITION_X/_Y/_Z	Cartesian X, Y, Z coordinate of interactive vehicle rear axle center position in the terrain database referential.
DISTANCE	Distance driven in [m]
SPEED	Current speed of the interactive vehicle [km/h]
ACCELERATION	Current acceleration of interactive vehicle in [m/s ²]
ACCELERATION_X/_Y/_Z	Current acceleration vector (X, Y, Z) in [m/s ²]
JERK	Current jerk of interactive vehicle in [m/s ³]
ACCELERATION_PEDAL	Position of the accelerator pedal of the interactive vehicle, between 0 (released) and 1 (fully pressed).
BRAKE_PEDAL	Position of the brake pedal of the interactive vehicle, in N/400.
CLUTCH_PEDAL	Position of the clutch pedal, between 0 (released) and 1 (fully pressed).
GEARBOX	Physical position of the gearbox level.
STEERING_WHEEL	Vehicle steering wheel position, between minimal radial and maximum radial. The position is 0 when the steering wheel is centered.
NEXT_VEHICLE_ID	Identifier of the first vehicle preceding the interactive vehicle on the studied lane.
DISTANCE_TO_NEXT_VEHICLE	Distance between interactive vehicle and vehicle on the studied lane.
LANE_LATERAL_SHIFT_CENTER_LEFT/_RIGHT	Lateral shift of the interactive vehicle from the center/left side/right side of the lane in [m].
LANE	Number associated with the lane on which the interactive vehicle is driving.
CONSUMPTION	Fuel consumption in [ml/s]
ALL_CONSUMPTION	Total fuel consumption in [ml]
INDICATORS	Indicators command of the vehicle, 1 for left, 2 for right indicator
WIDTH, HEIGHT, LENGTH	Width, height, and length of the vehicle in [m]
WHEEL_BASE	Distance between rear and front wheel in [m]
HEADING	The value of the specified angle coordinate of the angular position vector of the vehicle rear axle center. Heading angle specifies the vehicle rotation around its vertical axis. Heading is 0 if the vehicle heading is equal to the x-axis of the terrain coordinate system.
PITCH	Pitch angle specifies the <i>vehicle</i> rotation around its traversal axis. Pitch angle is positive if the front of the vehicle is lower than the rear. Pitch angle is 0 if the front of the car is as high as the rear.
ROLL	Roll angle specifies the <i>vehicle</i> rotation around its longitudinal axis. Roll is positive if the left side of the vehicle is above the right side. Roll angle is 0 if the left side of the car is as high as the right side.
REAL_AXEL_DISTANCE	Distance from real axle to the rear of the car in [m].
TIME_HEADWAY	Time between the real axle of the interactive car and real axle of the car in front of interactive car in seconds. If it empty, mean there were no car in front of interactive car.
COLLISION_ENTITY_TYPE	Object collision type: can be VEHICLE_VEHICLE, VEHICLE_OBJECT, OBJECT_VEHICLE, OBJECT_OBJECT, VEHICLE_AIRPLANESAFETYAREA, AIRPLANESAFETYAREA_VEHICLE, OBJECT_AIRPLANESAFETYAREA, AIRPLANESAFETYAREA_OBJECT, AIRPLANESAFETYAREA_AIRPLANESAFETYAREA
COLLISION_ID_1/_2	ID of the collided entity with interactive vehicle. Empty column means that there is no collision.
COLLISION_FORCE_X/_Y/_Z	The X/Y/Z component of the force of collision in the interactive vehicle, relation to the interactive vehicle. Empty column means that there is no collision.
COLLISION_LOCATION_X/_Y/_Z	The X/Y/Z component of the location of collision in the interactive vehicle, relation to the interactive vehicle. Empty column means that there is no collision.
COLLISION_ANGLE	Angle of the collision in the interactive vehicle, relation to the interactive vehicle. Empty column means that there is no collision.
ROAD PARAMETERS	
SPEED_LIMIT	Legal speed limit of the line in which the vehicle is driving
ROAD_SLOPE	Slope of the road, in % on the location of the real axle of the vehicle. If the value is zero, the road is flat. Value is positive when the road goes up and negative when the road goes down, according to the direction of the vehicle.

(continued on next page)

Table 4 (continued)

Parameter	Description
CURVE_RADIUS	Curve radius of the road, on the location of the real axle of the interactive vehicle. When the curve goes towards left according to the interactive vehicle, the value of curve radius is positive, when the curve goes towards right, the value is negative. Value 99000 represents a straight road.
ROAD_MARKING	Road marking of the interactive vehicle lane, in the left side of the vehicle. 0 means solid line, 1 means dotted line, 2 means no marking, -1 not working properly.
GET_LANE_TYPE_VEHICLE	Get the lane type on which the vehicle is running. The current lane of the vehicle is the lane on the middle of the vehicle rear axle [-30.16] or {vehicle not found, no lane, no road/unknown road, paved, paved entry, paved exit, paved express, paved roundabout, sidewalk, emergency, parking, zebra, median, shoulder, embankment, ditch, barrier, land, runaway, taxiway}.
LANE_WIDTH	Width of the lane at the point of vehicle in [m].
WEATHER	
RAIN	Rain amount coefficient ranging from 0 (no rain) to 1 (heavy rain).
WATER_ON_ROAD	Current water accumulation on road, from 0 to 20 mm.
SNOW	Current snow coefficient, from 0 (no snow) to 1 (strong snow).
FOG	Current visibility distance caused by fog in [m].
DAY_TIME	Current hour of the simulated day, from 0 to 24 h, with a floating-point value for minutes and seconds. 0 is midnight, 12 is midday, 14.5 is 14h30.
EMPATICA DATA	
ACCELERATION_X/_Y/_Z	Accelerometer that measures continuous gravitational force (g) applied to each of the three dimensions (x, y, and z). The scale is limited to $\pm 2g$.
BVP	The PPG (Photoplethysmography) data or BVP (Blood Volume Pulse) [nW]. The BVP is the input signal to the algorithm that detects the heartbeat.
GSR	Galvanic Skin Response represents change in the electrical resistance of the skin caused by emotional arousal, measurable with a sensitive galvanometer in [μS].
IBI	Interbeat interval represents interval between individual heart beats in [s/beat]. Heart rate can be calculated from this signal using the equation $HB = 60/IBI$ [beats/min].
ST	Skin temperature in [$^{\circ}C$].
TOBII DATA	
PUPIL_DIAMETER_LEFT/_RIGHT	The pupil diameter data is provided for the left and the right eye individually and is an estimate of the pupil size in millimeters.
T_PC_EYE	The pupil center is specified in 3D coordinates with origin in the scene camera that is integrated in the Head Unit.
T_PD_EYE	Pupil diameter for specific eye (1-right 2-left).
T_PD_DIAMETER	The pupil diameter is measured in mm.
TOBII-GAZE-X/_Y	The gaze position is the position on the scene camera image where the gaze will be projected. Top left corner is (0, 0), bottom right corner is (1,1).
TOBII-GAZE-SCREEN	Timestamp position on the screen (1-left, 2-center, 3-right), calculated by a proprietary module.
TOBII-GAZE-TIMESTAMP	Timestamp for gaze position.

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